PORTFOLIO 2026 · SHEET 00 / 07

Benjamin Harland

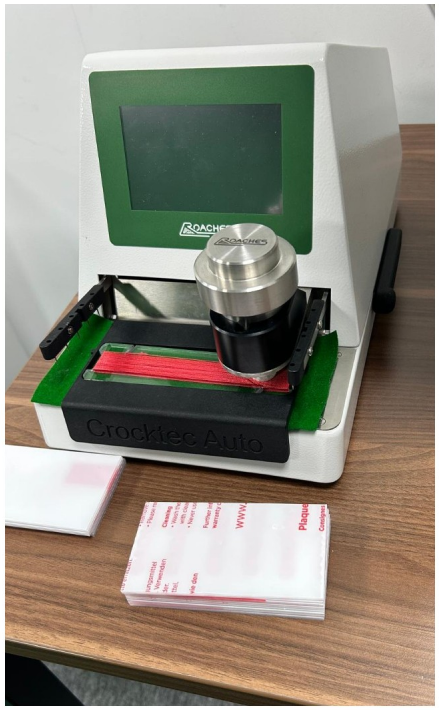
Mechanical design engineer

First Class BEng (Hons) in Mechanical Engineering from Liverpool John Moores University, 14 months as a Design Engineer at Roaches International, and now studying for an MSc in Advanced Mechanical Engineering at the University of Leeds.

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Winch dyeing machine

ROACHES INTERNATIONAL · 2024–25

Modernising a 1980s winch dyeing machine

An order for ten winch dyeing machines meant bringing a 1980s design up to date. I built the full CAD assembly from the original hand drawings, helped redesign several of the mechanisms and verified the design with FEA.

WHAT CHANGED

The tank sump was reshaped so the machine runs on less dye liquor, which has to be exchanged between batches of cloth. The mechanical systems and automation were updated to make setup less labour-intensive, with a new system for loading and unloading cloth and an updated control panel.

WHY IT MATTERS

The liquor is exchanged for every batch of cloth, so a smaller sump means less water, dye and chemicals per batch, and less liquor to heat. Simpler setup and loading also cut the operator time each batch takes.

ROACHES INTERNATIONAL · 2024–25

Redesigning two machines for an international exhibition

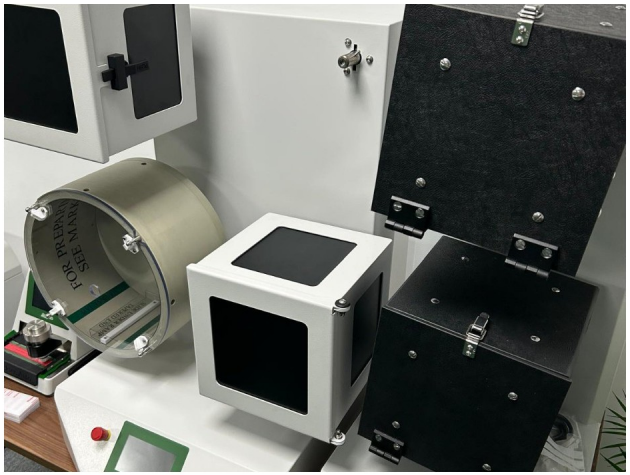
Both redesigns meant designing to each process's limits: turned and milled parts, laser-cut and folded sheet metal and its thickness, welded fabrications and 3D-printed parts, with attention to bearing selection, tolerances and fits, and what drives up manufacturing cost.

The aim was to make both machines easier to manufacture and assemble, and to fix customers' pain points, such as ease of use. I redesigned their drives and fabrications, ran the design meetings, and delivered both as working prototypes to a tight deadline.



Crocktec Auto

Tests colour fastness to rubbing. I designed a wire holster to keep wires from snagging on moving parts. The test standard sets the load on the sample, and tolerance stack-up calculations showed the design stays within it when parts are made to drawing.



OptiPill

Tests fabric pilling. Belt slip was knocking the boxes out of line, so I redesigned the whole drive: shafts, bearing housings, belt layout and main mounting plate. I also simplified the costly fabrication and gave it a slicker box style.



OptiPill: box mounting

A spring lock holds each box on the driven shaft so boxes go on and off quickly, and a braked motor stops the shaft spinning while one is fitted.

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Sample holders for the Sentire

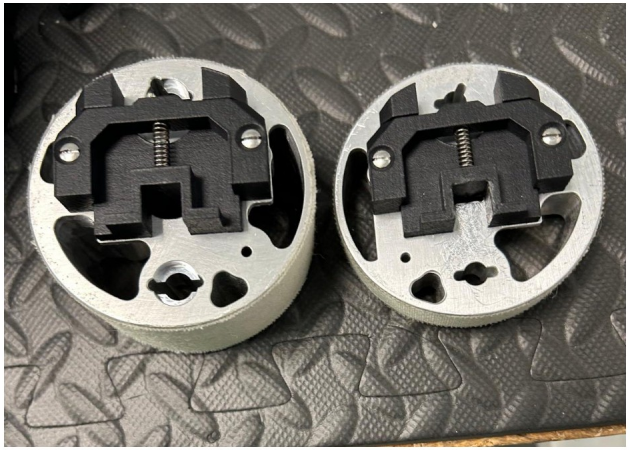
The Sentire tests fabric quality with high precision, so results only compare if every sample is treated the same. I designed the holders so all fabric is handled in exactly the same way, keeping tests consistent and reducing human error.

DESIGN

That control is why the bodies are machined from aluminium. I optimised them to reduce weight while keeping the centre of mass central, and paired them with 3D-printed clamps and rings.



Lightened bodies, centre of mass kept central



3D-printed clamps fitted



3D-printed rings

LIVERPOOL JOHN MOORES UNIVERSITY · 86%

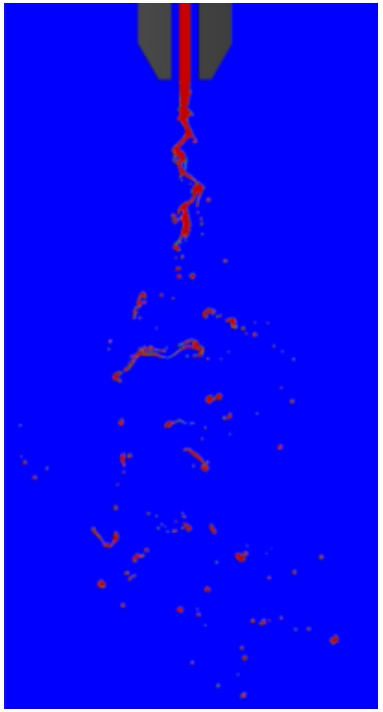
Atomisation in air-assisted nozzles

A Parametric Computational Fluid Dynamics Study of Newtonian and Non-Newtonian Fluid Atomisation in Air-Assisted Nozzles

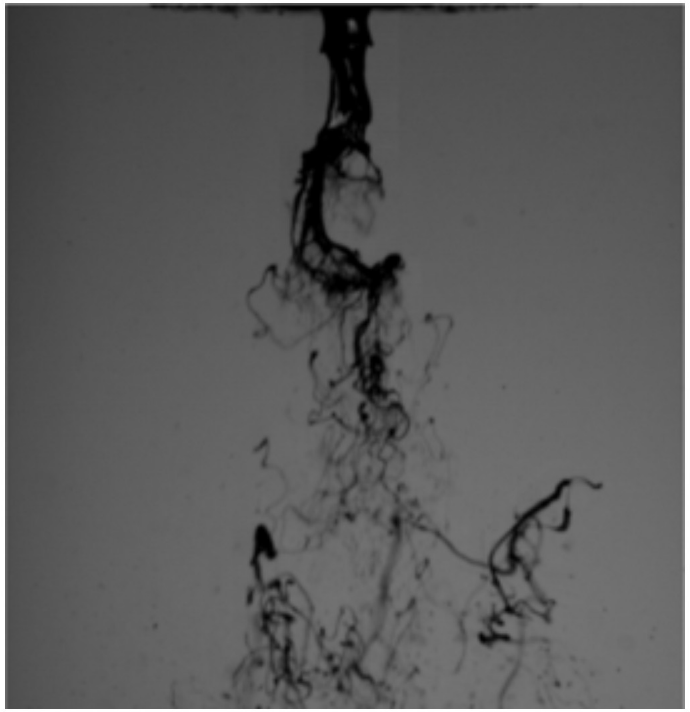
MODEL A 2D transient multiphase model in ANSYS Fluent, using the Volume of Fluid (VOF) method, RANS turbulence modelling and the Cross model for shear-thinning viscosity.

FINDINGS Under identical operating conditions, the shear-thinning fluid showed greater deformation, breakup and jet dispersion than the Newtonian fluid.

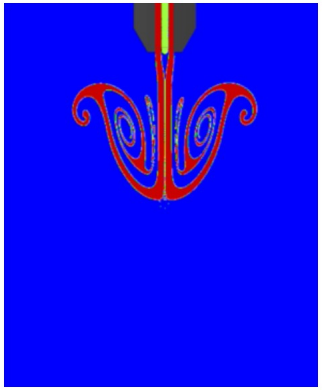
VALIDATION Newtonian results showed reasonable agreement with published experimental data (Jakobs et al., 2014). I traced instability in the non-Newtonian case to Courant number fluctuations.



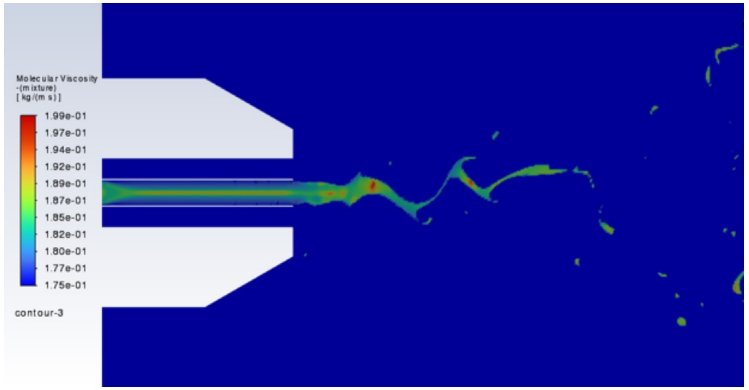
Newtonian, liquid VOF



Experiment: Jakobs et al. (2014)



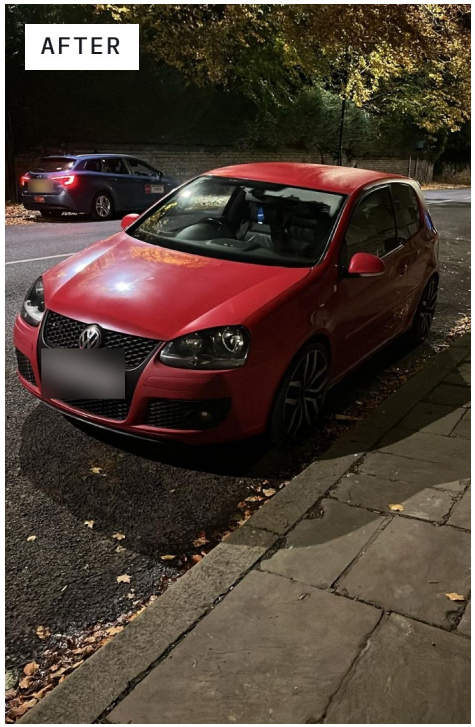
Newtonian, air VOF



Viscosity contour, Case 2: non-Newtonian







PERSONAL PROJECT

Golf Mk5 GTI restoration

I brought a non-running Golf GTI back to full running order: I resolved ECU, injector and spark plug problems, cleaned the intake manifold, and traced and replaced faulty sensors. I also rebuilt the front and rear suspension with new arms, bushings, struts and springs, and learnt basic MIG welding to repair the subframe.



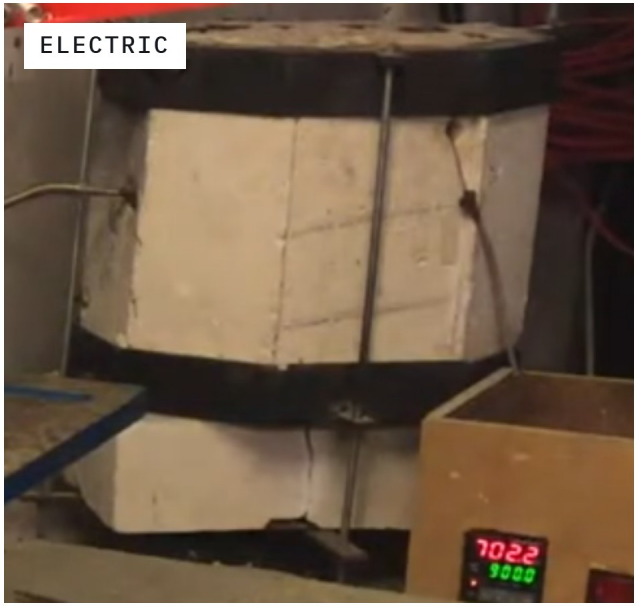


PERSONAL PROJECT

Citroën C1 frame repair

A friend's Citroën C1 had rusted through at a subframe mounting point. The fuel tank, suspension and subframe were removed for access, the corroded metal was cut back to sound steel, and a new sheet-steel section was welded in before reassembly.







PERSONAL PROJECT

Aluminium melting furnace

A furnace for melting down scrap aluminium and foil for sand casting. The first version was coal-fired: a cut-down gas cylinder held the fuel bed, a hairdryer acted as a forced-air blower to raise the combustion temperature, and the crucible sat directly in the coals. The second version switched to electric heating, with a chamber of insulating firebrick clamped by steel bands and tie rods, giving tighter temperature control and cleaner, easier running.

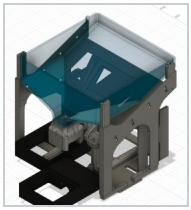
PERSONAL PROJECT

Hydraulic apple press

A 6-tonne hydraulic shop press, converted for cider-making. The jack drives a plate down into a perforated stainless steel basket, crushing the apples, while the pan beneath collects the juice. The conversion was fabricated with a mix of welded and bolted joints.

PERSONAL PROJECT

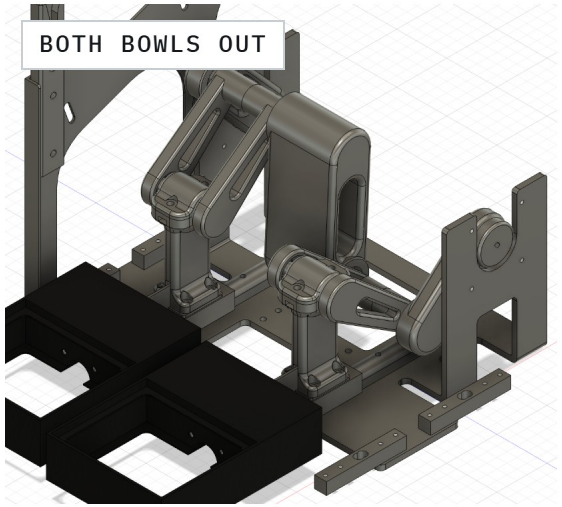
Dual cat feeder



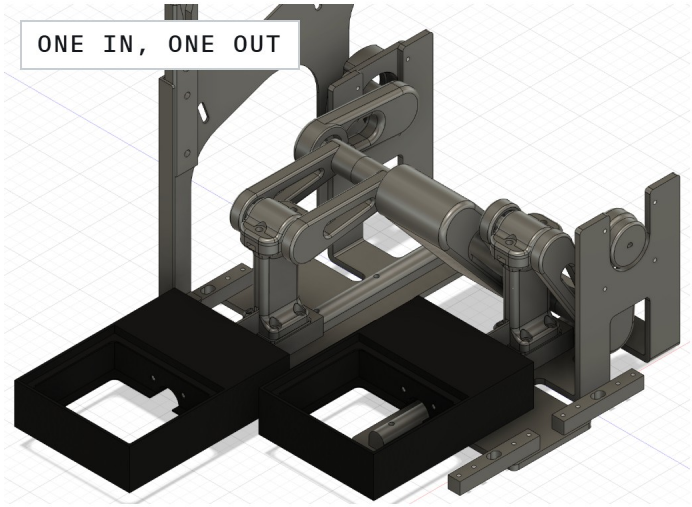
FULL ASSEMBLY

A feeder for two cats that reads the chip on each cat's collar and retracts the other cat's bowl, so each cat gets its own food. It will also dispense precise portions automatically throughout the day. A single motor drives both bowls: a crankshaft and connecting rods, as in an engine, convert its rotation into the linear motion of each bowl along a slider rail.

I'm currently designing the mechanical system, with a CAD model and 3D-printed parts so far, and will then move on to the electrical system, integrating the collar-chip reader with the motor control and planning a Zigbee link into Home Assistant so the feeder can be controlled from a phone.



BOTH BOWLS OUT



ONE IN, ONE OUT

Home automation

I set up and programmed a Home Assistant system, writing custom automations that link sensors, lighting, automatic curtains, central heating and cameras.

Arduino and electronics

I'm experimenting with Arduino kits and small circuits, wiring up buttons, switches and daylight sensors. No set project yet.

3D printing

I use my 3D printer for functional parts, such as a vacuum nozzle shaped to recover debris from inside the GTI's engine, a spring-loaded scraper for cleaning stovetop espresso pots and fixings to hold a lamp's bulb in place, as well as Dungeons & Dragons figurines.